ACS APPLIED MATERIALS & INTERFACES

www.acsami.org

Research Article

Homogenizing Electron and Hole Transport Layers in Tin Perovskite Solar Cells to Enhance Photocurrent and Voltage

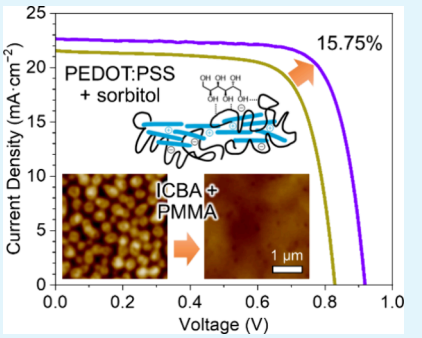
Qinyu Ning,[†] Donghao Miao,[†] Mingyu Ma, Zihao Zhu, Yi Chen, Yuchen Ding, Wenzhuo Li, Wei Zhou, Yuedong Shi, Zhijun Ning, and Qixi Mi*

Cite This: https://doi.org/10.1021/acsami.5c05263

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Tin perovskite solar cells (TPSCs) are considered as a sustainable alternative to their lead-containing counterparts, offering an ideal bandgap, a more stable crystal structure, and better semiconducting properties. However, the fabrication of efficient and able TPSCs has been a challenging task. Among the main components of TPSCs, herein, we investigated the transport layers and their interface with the perovskite layer. Several microscopy techniques revealed inhomogeneity in the common hole transport layer, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS), and the electron transport layer (ETL), indene-C_{60} bisadduct (ICBA). A hydrogen-bond donor molecule D-sorbitol modified the microstructure of PEDOT:PSS, increasing its conductivity and the crystallization density of the overlying perovskite layer. Blending 1% poly(methyl methacrylate) into ICBA rendered a uniform ETL, which facilitated electron transport across its interface with tin perovskite. The photocurrent density, open-circuit voltage, and fill factor of the TPSCs were all enhanced to achieve a power conversion efficiency of 15.8% and excellent short-term stability. Our findings address an often-neglected aspect of device fabrication and provide a new approach to boosting the performance and reproducibility of TPSCs.



KEYWORDS: *tin perovskite, charge transport layer, PEDOT:PSS, ICBA, efficiency*

Downloaded via SHANGHAITECH UNIV on May 24, 2025 at 03:55:50 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

INTRODUCTION

Organic−inorganic metal halide perovskites (MHPs) have been receiving increasing attention thanks to their low-cost raw materials and convenient manufacturing process. Over the past decade, the power conversion efficiency (PCE) of single junction perovskite solar cells (PSCs) has increased from 3.8 to over 26%,[1−6] while all-perovskite and silicon-perovskite tandem solar cells have reached 30 and 34% respectively.[7−11] However, concerns about the toxicity of lead have hampered their extensive application. Tin perovskite solar cells (TPSCs) are regarded as the most promising substitution to lead and have superior optoelectronic properties such as an ideal bandgap (E_g) of 1.4 eV, lower exciton binding energy, and higher carrier mobility. Therefore, Sn perovskites are ideal materials for indoor photovoltaics, consumer electronics, and flexible devices.

During the last year, tin-based perovskite solar cells have achieved a PCE of ∼16%,[12−14] which is still far below the Shockley−Queisser (SQ) limit. Great efforts have been made to address the issue that Sn^{2+} is more easily oxidized than Pb^{2+}. Antioxidant additives, such as SnF_2, hypophosphorous acid, vitamin C, and hydrazine derivatives,[15−18] have been proposed to prevent the oxidation of Sn^{2+}. Thiocyanate (SCN^-), tetrafluoroborate (BF_4^-), and trimethylthiourea (3T) have been shown[19−21] to be effective ligands to Sn^{2+} and regulate the crystallization of tin perovskite films. Organic ammonium additives such as 4-fluorophenethylammonium bromide (FPEABr) can induce a mixed 2D/3D crystal structure in TPSCs[22] and have become a commonly adopted strategy.

For TPSCs in the inverted p-i-n structure, the interfacial contact between the perovskite and charge transport layers is another important issue. Regarding the hole transport layer (HTL), poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is widely used for its high optical transparency and suitable energy level.[23,24] Nevertheless, the coiled PSS polymer chains in PEDOT:PSS can lead to insufficient contact with the bottom electrode and restrict the current conduction. Han et al. intercalated polyethylene glycol (PEG) into PEDOT:PSS and reduced the energy-level mismatch between $FASnI_3$ and PEDOT:PSS, producing reduced hysteresis and improved stability.[25] Several other studies[26−29] demonstrated that introducing small, polar molecules can induce conformational change in the PSS polymer chain, thereby altering its structure.

Received: March 20, 2025
Revised: May 15, 2025
Accepted: May 16, 2025

© XXXX American Chemical Society

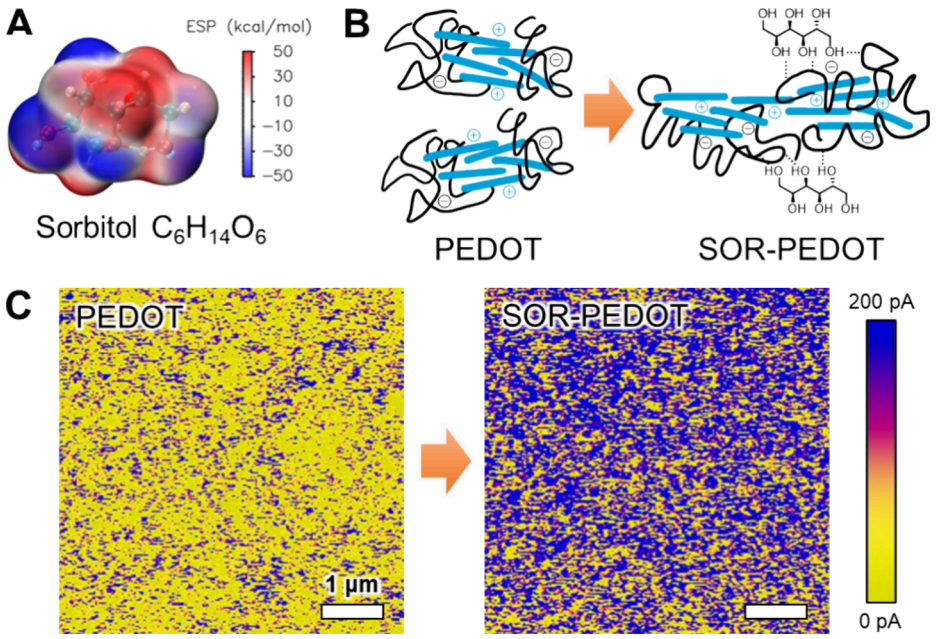


Figure 1. (A) Electrostatic potential (ESP) map of D-sorbitol (SOR). (B) Schematic diagram of the PEDOT:PSS polymer backbone modified by SOR. (C) Conductive atomic force microscopy (c-AFM) of PEDOT and SOR-PEDOT films.

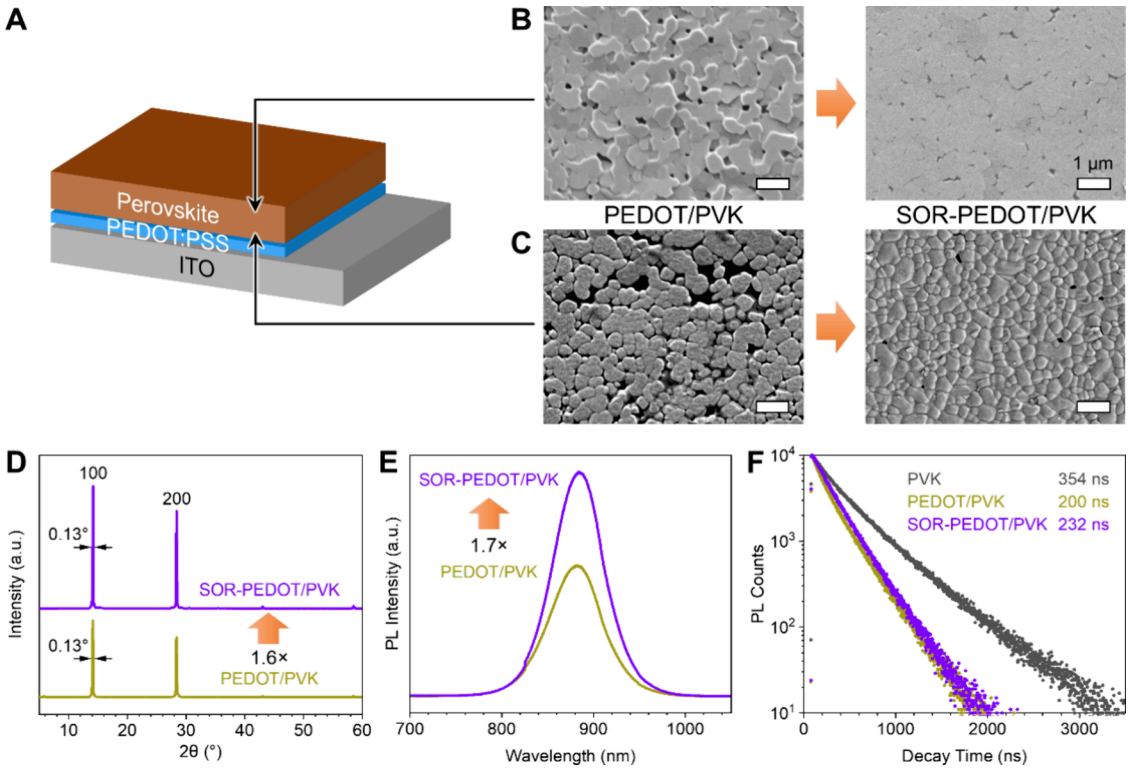


Figure 2. (A–C) Scanning electron microscopy (SEM) images of the top and buried surfaces of the perovskite layer deposited on PEDOT or SOR-PEDOT substrates. (D–F) XRD pattern, photoluminescence (PL) spectra, and decay kinetics of the perovskite layer deposited on PEDOT or SOR-PEDOT substrates.

For the electron transport layer (ETL), Ning et al.[30] replaced phenyl-C_{61}-butyric acid methyl ester (PCBM) with indene-C_{60} bisadduct (ICBA) as the electron acceptor material and raised the photovoltage to 0.94 V. Liao et al.[31] reported a post-treatment method using 3-aminomethylbenzo[*b*]-thiophene (3-AMBTh), which enhanced $\pi-\pi$ stacking interactions with ICBA. Liang et al.[32] deposited ICBA on a surface-anchored n-type polymer poly(naphthalene diimide-*alt*-dithiophenebenzothiadiazole) (PNDI-BT), which exhibits high electron mobility and excellent stability. Despite these significant studies, ETLs made of ICBA usually exhibit poor reproducibility in device performance, and the reason remains unclear.

https://doi.org/10.1021/acsami.5c05263
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

In this work, we investigate the origin of inhomogeneity in the electron and hole transport layers for tin perovskites and utilize additives to achieve a more even morphology in these transport layers. Specifically, D-sorbitol, a sugar alcohol, modified the microstructure of PEDOT:PSS, which improved the crystallization of the overlying tin perovskite layer. On the other hand, poly(methyl methacrylate) (PMMA) prevented ICBA from aggregating during solution processing, giving rise to a uniform coverage of the tin perovskite layer. The homogeneous electron and hole transport layers significantly improved the photovoltaic performance of TPSCs.

RESULTS AND DISCUSSION

Figure 1A depicts the electrostatic potential (ESP) map of the natural product D-sorbitol (SOR), calculated using density functional theory (DFT). The chemical structure of SOR with six hydroxyl groups provides multiple hydrogen bonding sites, indicated by the noticeable positive ESP (red regions). In the commercial formulation of PEDOT:PSS used herein, one part of PEDOT is surrounded by approximately six parts of the PSS polyanion. The nonconducting PSS surfactant not only helps PEDOT form colloidal micelles but also hinders charge carrier transport between individual PEDOT domains. D-Sorbitol molecules are expected to form hydrogen bonds with the anionic sulfonate groups in PSS and partly disentangle the coiled PSS chains, which allows better overlap of the conducting PEDOT domains (Figure 1B). We further performed conductive atomic force microscopy (c-AFM) to investigate the effect of D-sorbitol doping on PEDOT:PSS. Figure 1C shows that a pristine PEDOT film conducts electrical current only in scattered patches, but in an SOR-PEDOT film, more than half the covered area is conducting, indicating that the latter sample is more homogeneous and electrically conducting.

Figure 2 characterizes the tin perovskite layer deposited on PEDOT or SOR-PEDOT substrates using scanning electron microscopy (SEM), X-ray diffraction (XRD), and photoluminescence (PL). In Figure 2B,C, both the top and buried surfaces of the perovskite layer became denser and more compact when deposited on the SOR-PEDOT substrate (denoted as SOR-PEDOT/PVK hereinafter). Analysis of these SEM images indicates that the area coverage of the PEDOT/PVK interface increased from 84 to 99% after adding D-sorbitol (Figure 2C). XRD peaks of the SOR-PEDOT/PVK sample, particularly the 100 and 200 peaks of $FASnI_3$, are systematically stronger than those of PEDOT/PVK by a factor of 1.6. The PL spectra of these two samples exhibit a similar trend in the signal intensity. On the other hand, the diffraction peak width and PL lifetime, which reflect the quality of perovskite crystallization, are comparable for these two samples. Therefore, results presented in Figure 2D−F also suggest that incorporating D-sorbitol in PEDOT enhanced the coverage density of the perovskite layer on top of it, without affecting the quality of individual perovskite crystallites.

Figure 3A illustrates the p-i-n solar cell architecture by using both a cross-sectional SEM image and a structural diagram. The choice of material for the n-type electron transport layer on top of tin perovskite is debated. Fullerene C_{60} and its monoadduct derivative PCBM possess advantageous charge transport and film-forming properties,[33−37] but the energy levels of these two electron acceptors are not high enough to match that of Sn perovskite, which induces losses in the open-circuit voltage (V_{OC}). ICBA, a fullerene bisadduct, aligns better

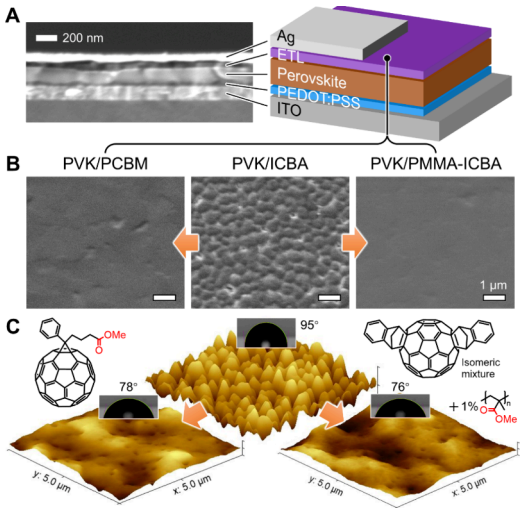


Figure 3. (A) Cross-sectional SEM image and device structure of tin perovskite solar cells. (B) Top-view SEM images of spin-coated PCBM, ICBA, and PMMA-ICBA films on the perovskite layer. (C) 3D height images of the three films in panel (B) measured by atomic force microscopy (AFM). Insets: water contact angles of these films and molecular structures of PCBM, ICBA, and PMMA.

with the energy level of tin perovskites (Figure S1) but gives poor reproducibility in film deposition and device performance. In this work, we synthesized and purified ICBA in a relatively high yield to ensure its quality. Nonetheless, spin-coating a filtered chloroform solution of ICBA produced an ETL laden with aggregates, evident to the naked eye (Figure S2). At the microscopic scale, scanning electron microscopy (SEM) and atomic force microscopy (AFM) in Figure 3B reveal that the ICBA film spin-coated on tin perovskite is not continuous but composed of discrete, granular aggregates of 0.27 ± 0.07 μm in diameter.

We identified PMMA, a common linear polymer, as an active ingredient to achieve the homogeneous deposition of ICBA from solution. Figure S3 shows characteristic infrared stretching vibrations of the ester group to confirm the existence of PMMA in its mixture with ICBA at a low mass ratio of $x \approx 1\%$. Introducing PMMA effectively converted the spin-coated ICBA to a smooth and uniform film, having a morphology similar to PCBM, although at a much lower content of the ester group. Both PMMA-ICBA and PCBM films exhibit an AFM roughness of ∼4 nm, in contrast to ∼14 nm for pure ICBA (Figure 3).

We propose that the polar C═O bond in PMMA, just like that in PCBM, can coordinate to Sn^{2+} on the perovskite surface, thereby creating an adherent electron transport layer on the perovskite. Otherwise, the ICBA precursor solution in chlorobenzene does not wet the perovskite surface effectively so as to form (sub)microscopic droplets during evaporation and then aggregates after annealing. Another similarity between PMMA-ICBA and PCBM films lies in their water contact angles of 78 and 76°, respectively, while that of a pure, agglomerated ICBA film is much larger at 95°. Now that the chemical composition difference between PCBM and PMMA-ICBA does not result in an appreciable change in their water contact angles, it is likely that the large water contact angle on pure ICBA is due to a structural effect, reminiscent of the hydrophobic surface of lotus leaves.

https://doi.org/10.1021/acsami.5c05263
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

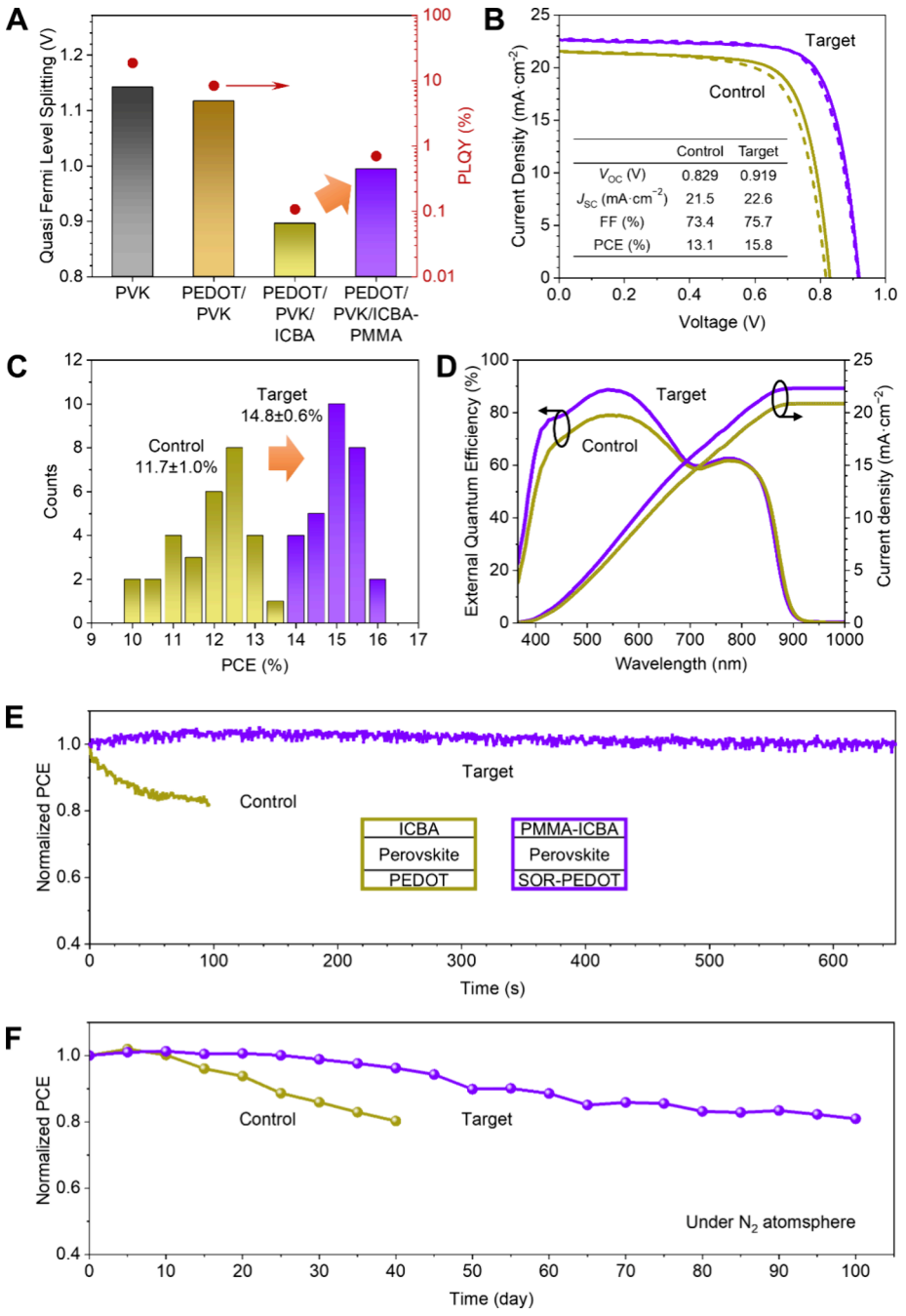


Figure 4. (A) PL quantum yield and implied open-circuit voltage (iV_{OC}) of tin perovskite in contact with different transport layers. (B–D) Photocurrent density–voltage scans, power conversion efficiency (PCE) statistics, and external quantum efficiency of the TPSC devices. (E, F) Short- and long-term performance stability data under simulated sunlight and a N_2 atmosphere.

The more complete and uniform deposition of PMMA-ICBA on top of tin perovskite has consequences for its electrical properties. Figure S4 plots the contact potential difference (CPD) of the three ETLs measured by Kelvin probe force microscopy (KPFM). Under comparable experimental conditions, PCBM, ICBA, and PMMA-ICBA show a progressively more negative CPD, which suggests more effective electron harvesting across the PVK/ETL interface in order. The mixing ratio x of PMMA in ICBA was optimized in a series of gradient experiments. In Figure S5, when the PMMA content was increased from 0 to 1.5%, the open-circuit voltage (V_{OC}) of the solar cell device grew monotonously. However, the photocurrent density (J_{SC}) maximized at x = 1.0% and abruptly dropped at x = 1.5%, also bringing down the fill factor (FF) and power conversion efficiency (PCE). This result is consistent with the electrical isolation of PMMA, and hereinafter, x = 1.0% was adopted in the target solar cell devices for convenience.

Next, the performance of TPSCs consisting of homogeneous ETL and HTL was characterized. In Figure 4A, nonradiative recombination losses at the HTL/PVK and PVK/ETL interfaces were quantified by photoluminescence quantum yield (PLQY) measurements. Tin perovskite by itself on glass shows a high PLQY, corresponding to an implied open-circuit voltage (iV_{OC}) of 1.14 V, close to the Shockley–Queisser limit for this material.[38] Spin-coating tin perovskites on PEDOT or

https://doi.org/10.1021/acsami.5c05263
ACS Appl. Mater. Interfaces XXXX, XXX, XXX–XXX

SOR-PEDOT substrates leads to only a mild reduction in PL lifetime (Figure 2F), PLQY, or iV_{OC}. By contrast, a discontinuous layer of pure ICBA on tin perovskite drastically cuts the iV_{OC} by >0.2 V, which was ameliorated by switching to a homogeneous layer of PMMA-ICBA. Coordination of PMMA's carbonyl group to Sn^{2+}, which aids in interfacial connection, could also contribute to the raised iV_{OC} of the target device.[39−41]

The effect of our strategy on the photovoltaic performance of the TPSCs was examined under AM1.5G illumination. In Figure 4B, the V_{OC} of a champion target device containing homogeneous ETL and HTL is 90 mV higher than the control device, parallel to the trend in their iV_{OC} values. J_{SC} is also boosted from 21.5 mA·cm^{-2} for the target device to 22.6 mA·cm^{-2} for the target device, attributable to more facile charge carrier extraction across the top and buried interfaces of the tin perovskite. Accompanied by an increased FF, the power conversion efficiency of the champion cell achieved 15.8%, along with reduced hysteresis and narrower data distribution (Figure 4C). Employing PCBM as the ETL also realized a J_{SC} of 22.6 mA·cm^{-2} at the expense of V_{OC} owing to energy-level mismatch with tin perovskite (Table S2). External quantum efficiency (EQE) measurements corroborate that charge carrier extraction becomes more efficient for the target group, while the light absorption features of the perovskite layer remain unaffected (Figure 4D). Short-term performance stability was tracked at the maximum power point (MPP). Throughout the first testing period of 600 s, the target device exhibited essentially no degradation in its PCE, while the control device decayed to 80% of its initial performance within 100 s (Figure 4E). The shelf life of the target devices was >60 days under a N_2 atmosphere, more than twice that of the control devices (Figure 4F). Overall, the target device features a comprehensive upgrade in all the key performance parameters.

CONCLUSIONS

To summarize, previous studies on TPSCs have provided access to high-quality tin perovskite layers and compatible electron/hole transport materials, and our work tackles a problem that is usually overlooked: the homogeneity of solution-processed ETL and HTL. D-Sorbitol reconfigured the PEDOT:PSS backbone, boosted its hole conductivity, and benefited the crystallization of the overlying tin perovskite. Incorporation of 1% PMMA in ICBA reshaped its morphology from isolated aggregates to a uniform film, which suppressed interfacial losses and enhanced V_{OC} and J_{SC} values simultaneously. As a result, target devices attained a champion PCE of 15.8% and excellent short-term stability, among the highest performance for TPSCs reported recently (Table S3). D-Sorbitol and PMMA are low-cost materials, and their use as additives to the precursor solution is fully compatible with the solution processing procedure for industrial applications. This work emphasizes the importance of microstructures (morphology, texture, surface properties, etc.) for organic transport materials, more than just energy-level matching, and provides insights into developing next-generation transport materials (such as nonfullerene acceptors) for higher-performance and more scalable TPSCs.

EXPERIMENTAL SECTION

Materials. All tin(II) compounds were stored and handled in N_2 gloveboxes (Vigor Tech) containing <0.1 ppm of O_2 and H_2O at 18−22 °C. *N*,*N*-Dimethylformamide (DMF, 99.8%, anhydrous), dimethyl sulfoxide (DMSO, 99.9%, anhydrous), and stannous fluoride (SnF_2, 99%) were obtained from J&K Scientific. Chlorobenzene (99.8%, anhydrous) was obtained from Sigma-Aldrich. NH_4SCN (99.99% metal basis) and bathocuproine (BCP, 98%) were obtained from Aladdin. Methanol (99.7%), tin granules (5 N), and iodine (99.8%) were obtained from Sinopharm. *N*-Methylthiourea (99%), trimethylthiourea (98%), D-sorbitol (99.5%), and poly(methyl methacrylate) (PMMA, average M_w ∼100,000) were obtained from Macklin. Formamidinium iodide (FAI) and indene (98%) were obtained from Tokyo Chemical Industry. The above raw materials were used without further purification. Toluene (99.5%) was distilled in the presence of a Na−K alloy and benzophenone ketyl inside the N_2 glovebox. Thiophene-2-ethylamine hydroiodide (TEAI) was from our previous work.[42]

PEDOT:PSS. PEDOT:PSS (Clevious P VP AI 4083, ∼1% aqueous dispersion) was added D-sorbitol at a concentration of 10 mg/mL. The mixture was stirred at room temperature and then passed through a 0.22 μm PVDF filter before use.

Indene-C_{60} Bisadduct (ICBA). Indene-C_{60} bisadduct (ICBA) was synthesized from C_{60} (99.96%, Suzhou Tanfeng) and indene according to previous reports[43,44] and purified by flash column chromatography (Santai SepaBean T) as a brownish black solid. Yield: 51%. MALDI-TOF MS (dithranol): *m*/*z* calculated for $C_{78}H_{17}^+$ ($[M + H]^+$) 953.13, found 953.17.

Tin Perovskite Precursor Solution. A 0.85 M SnI_2 solution was prepared by reacting I_2 with excess Sn granules in a 4:1 (v/v) mixture of DMF and DMSO according to our previous protocol.[21] Formamidinium iodide (FAI, 126 mg, 0.733 mmol), TEAI (34.7 mg, 0.134 mmol), *N*-methylthiourea (8.0 mg, 0.089 mmol), trimethylthiourea (40.0 mg, 0.338 mmol), SnF_2 (14.1 mg, 0.090 mmol), and NH_4SCN (5.0 mg, 0.066 mmol) were completely dissolved in 1.00 mL of the above SnI_2 solution at room temperature. The resulting clear, bright yellow solution was filtered through a 0.22 μm PTFE filter before use.

Device Fabrication and Evaluation. Indium tin oxide (ITO) conducting glass slides (Advanced Election Tech Co., Ltd., 8 Ω·sq^{-1}, 2.5 × 2.5 cm^{-2}) were cleaned using detergent/deionized water, acetone, and ethanol under ultrasonication and treated with ultraviolet light and ozone (SETCAS SC−UV-I) for 20 min. The PEDOT:PSS dispersion was spin-coated onto the ITO substrates at 5000 rpm for 45 s (Schwan Technology EasyCoater 6) and annealed at 140 °C for 20 min. Inside the N_2 glovebox, the perovskite precursor solution was spin-coated (Sawatec SM-150) on PEDOT:PSS substrates at 1000 rpm for 10 s and 5000 rpm for 30 s. Crystallization of the perovskite layer was promoted by a toluene rinse (550 μL) 10 s before the end of the spin, and then, the substrates were annealed at 70 °C for 10 min. A solution of ICBA (30 mg·mL^{-1}) and PMMA (0.3 mg·mL^{-1}) in chlorobenzene was spin-coated (Sawatec SM-150) on top of the ITO/PEDOT:PSS/perovskite substrates at 1500 rpm for 30 s, followed by annealing at 70 °C for 10 min. Subsequently, bathocuproine (BCP, 5 nm) and Ag electrodes (80 nm) were formed under a high vacuum in a thermal evaporator (Anhui Jiashuo JSD400) attached to the N_2 glovebox.

Current Density−Voltage (*J*−*V*). Current density−voltage (*J*−*V*) characteristics of these solar cells were measured between −0.1 and 1.1 V at steps of 0.01 V using a Keithley 2450 source/meter, under 100 mW·cm^{-2} for simulated AM1.5G illumination (Zolix Sirius-SS150A) through a 0.20 × 0.20 cm^2 mask.

External Quantum Efficiencies (EQE). External quantum efficiencies (EQE) were measured using a tunable monochromatic light source (Zolix TLS3-T150A), projected onto solar cells in the glovebox via a feed-through optical fiber. Under zero bias voltage, the wavelength-dependent photocurrents were monitored by a Keithley 2450 source/meter and referenced to a calibrated Si solar cell (Zolix QE-B3) with matching spectral range. Integrated photocurrent density was derived from experimental EQEs and the standard AM1.5G spectrum.

X-ray Diffraction (XRD). Powder XRD patterns of anealled perovskite films on ITO/PEDOT:PSS substrates were acquired on a

https://doi.org/10.1021/acsami.5c05263
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

Bruker D2 Phaser emitting Cu Kα radiation (λ = 1.5418Å) under ambient conditions.

Scanning Electron Microscopy (SEM). Top, bottom, and cross-sectional micrographs of tin perovskite films on different substrates were imaged by a field-emission scanning electron microscope (JEOL JSM-7800F) under an accelerating voltage of 2 kV and a working distance of 8 mm. Images were processed by ImageJ using its "Threshold" and "Analyze Particles" functions.

Atomic Force Microscopy (AFM). The top view of various samples of the hole or electron transport layer was imaged by Bruker Dimension Fastscan and the AFM data processed by the Nanoscope Analysis and Gwyddion software. Table S1 lists the experimental conditions for the normal AFM, c-AFM, and KPFM techniques.

The Fourier transform infrared spectra (FT-IR) of fullerene derivatives were acquired on a Bruker Vertex 70 spectrometer using liquid cells. Steady-state photoluminescence (PL) spectra were collected on a Horiba Fluorolog-3 fluorimeter by using 500 nm excitation and a CCD detector under ambient conditions. For time-resolved photoluminescence (TRPL), perovskite films were excited by a 510 nm picosecond diode laser in a lifetime spectrometer (Edinburgh Instruments Mini-Tau) in the N_2 glovebox, and the PL photons were detected by a red-extended photomultiplier tube (Hamamatsu H71020-20) and analyzed using the time-correlated single-photon counting (TCSPC) technique, fitted by a single-exponential decay model. Photoluminescence quantum yield (PLQY) measurements were performed using the LuQY Pro apparatus (Quantum Yield Berlin) in the N_2 glovebox, with a 515 nm laser providing incident light equivalent to 1 sun illumination. The PL spectra were quantified and converted to quasi-Fermi level splitting (QFLS) or implied open-circuit voltage (iV_{OC}) automatically.

Contact Angle. A water drop on the electron transport layer was pictured by an SL200 KS (KINO) contact angle meter in the Pendant Drop mode and fitted by the Laplace–Young method.

Computations. Gaussian 09 was used to model the molecular structure and electrostatic potential (ESP) of D-sorbitol using the B3LYP functional and the 6-31G basis set. The calculation results were visualized by using the VMD software.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications Web site at DOI: 10.1021/acsami.xxxxxx. The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.5c05263.

Energy band alignment diagram, supplementary characterizations of the ETL, performance statistics, and additional references (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Qixi Mi − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0000-0001-7644-1417; Email: miqx@shanghaitech.edu.cn

Authors

Qinyu Ning − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0009-0009-0455-3933

Donghao Miao − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0000-0003-1222-280X

Mingyu Ma − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China*

Zihao Zhu − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0000-0001-7793-3026

Yi Chen − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China*

Yuchen Ding − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China*

Wenzhuo Li − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China*

Wei Zhou − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China*

Yuedong Shi − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0009-0001-4331-3299

Zhijun Ning − *School of Physical Science and Technology and State Key Laboratory of Advanced Medical Materials and Devices, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0000-0002-9130-3490

Complete contact information is available at: https://pubs.acs.org/10.1021/acsami.5c05263

Author Contributions

†Q.N. and D.M. contributed equally to this work.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work is financially supported by the National Natural Science Foundation of China (22075182), the Shell–CAS Frontier Sciences Program (PT78963), and funding from ShanghaiTech University. The authors thank the Center for High-Resolution Electron Microscopy (EM02161943), Analytical Instrumentation Center and Soft Matter Nanofabrication Laboratory of ShanghaiTech University for instrumentation, Dr. Xinyan Wang for assistance with atomic force microscopy measurements and analysis, and Shanghai Coupletechs Co., Ltd. for lending the LuQY Pro apparatus.

■ REFERENCES

(1) Kojima, A.; Teshima, K.; Shirai, Y.; Miyasaka, T. Organometal Halide Perovskites as Visible-Light Sensitizers for Photovoltaic Cells. *J. Am. Chem. Soc.* **2009**, *131* (17), 6050–6051.

(2) Burschka, J.; Pellet, N.; Moon, S.-J.; Humphry-Baker, R.; Gao, P.; Nazeeruddin, M. K.; Grätzel, M. Sequential deposition as a route to high-performance perovskite-sensitized solar cells. *Nature* **2013**, *499* (7458), 316–319.

(3) Chen, H.; Liu, C.; Xu, J.; Maxwell, A.; Zhou, W.; Yang, Y.; Zhou, Q.; Bati, A. S. R.; Wan, H.; Wang, Z.; et al. Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* **2024**, *384* (6692), 189–193.

(4) Ge, Y.; Zheng, L.; Wang, H.; Gao, J.; Yao, F.; Wang, C.; Li, G.; Guan, H.; Wang, S.; Cui, H.; et al. Suppressing wide-angle light loss and non-radiative recombination for efficient perovskite solar cells. *Nat. Photonics* **2025**, *19* (2), 170–177.

(5) Zai, H.; Yang, P.; Su, J.; Yin, R.; Fan, R.; Wu, Y.; Zhu, X.; Ma, Y.; Zhou, T.; Zhou, W.; et al. Wafer-scale monolayer MoS2 film integration for stable, efficient perovskite solar cells. *Science* **2025**, *387* (6730), 186−192.
(6) Wang, D.; Liu, Z.; Qiao, Y.; Jiang, Z.; Zhu, P.; Zeng, J.; Peng, W.; Lian, Q.; Qu, G.; Xu, Y.; et al. Rigid molecules anchoring on NiOx enable > 26% efficiency perovskite solar cells. *Joule* **2025**, *9*, No. 101815.
(7) Chen, J.; Du, J.; Cai, J.; Ouyang, B.; Li, Z.; Wu, X.; Tian, C.; Sun, A.; Zhuang, R.; Wu, X.; et al. Lewis Base Strategy for Crystallization Control and Buried Interface Passivation on Hydrophobic PTAA Substrate for Efficient Tin−Lead Perovskite and All-Perovskite Tandem Solar Cells. *ACS Energy Lett.* **2025**, *10*, 1117−1128.
(8) Liu, Z.; Lin, R.; Wei, M.; Yin, M.; Wu, P.; Li, M.; Li, L.; Wang, Y.; Chen, G.; Carnevali, V.; et al. All-perovskite tandem solar cells achieving > 29% efficiency with improved (100) orientation in wide-bandgap perovskites. *Nat. Mater.* **2025**, *24* (2), 252−259.
(9) Fang, Z.; Deng, B.; Jin, Y.; Yang, L.; Chen, L.; Zhong, Y.; Feng, H.; Yin, Y.; Liu, K.; Li, Y.; Zhang, J.; Huang, J.; Zeng, Q.; Wang, H.; Yang, X.; Yang, J.; Tian, C.; Xie, L.; Wei, Z.; Xu, X.; et al. Surface reconstruction of wide-bandgap perovskites enables efficient perovskite/silicon tandem solar cells. *Nat. Commun.* **2024**, *15* (1), 10554.
(10) Yang, Y.; Chang, Q.; Su, J.; Chao, L.; Wang, Y.; Dai, Z.; Huang, X.; Nie, S.; Guo, P.; Yin, J.; et al. Activating Halogen Circulation Enables Efficient and Stable Wide-Bandgap Mixed-Halide Perovskite Solar Cells. *Adv. Mater.* **2025**, *37*, No. 2416513.
(11) Green, M. A.; Dunlop, E. D.; Yoshita, M.; Kopidakis, N.; Bothe, K.; Siefer, G.; Hao, X.; Jiang, J. Y. Solar Cell Efficiency Tables (Version 65). *Prog. Photovolt.: Res. Appl.* **2025**, *33* (1), 3−15.
(12) Shi, Y.; Zhu, Z.; Miao, D.; Ding, Y.; Mi, Q. Interfacial Dipoles Boost Open-Circuit Voltage of Tin Halide Perovskite Solar Cells. *ACS Energy Lett.* **2024**, *9* (4), 1895−1897.
(13) Chen, J.; Luo, J.; Li, Y.; Chen, X.; Song, Z.; Hou, E.; Sun, C.; Zhang, H.; Cheng, S.; Xing, Y.; Chen, S.; Zhao, X.; Xie, L.; Tian, C.; Wei, Z.; et al. Spatially Isomeric Fulleropyrrolidines Enable Controlled Stacking of Perovskite Colloids for High-Performance Tin-Based Perovskite Solar Cells. *Angew. Chem., Int. Ed.* **2025**, *64* (7), No. e202420150.
(14) López-Fernández, I.; Valli, D.; Wang, C.; Samanta, S.; Okamoto, T.; Huang, Y.; Sun, K.; Liu, Y.; Chirvony, V. S.; Patra, A.; Zito, J.; De Trizio, L.; Gaur, D.; Sun, H.; Xia, Z.; Li, X.; Zeng, H.; Mora-Seró, I.; Pradhan, N.; Martínez-Pastor, J. P.; Müller-Buschbaum, P.; Biju, V.; Debnath, T.; Saliba, M.; Debroye, E.; Hoye, R. L. Z.; Infante, I.; Manna, L.; Polavarapu, L.; et al. Lead-Free Halide Perovskite Materials and Optoelectronic Devices: Progress and Prospective. *Adv. Funct. Mater.* **2024**, *34* (6), 2307896.
(15) Gupta, S.; Cahen, D.; Hodes, G. How SnF2 Impacts the Material Properties of Lead-Free Tin Perovskites. *J. Phys. Chem. C* **2018**, *122* (25), 13926−13936.
(16) Wang, C.; Gu, F.; Zhao, Z.; Rao, H.; Qiu, Y.; Cai, Z.; Zhan, G.; Li, X.; Sun, B.; Yu, X.; Zhao, B.; Liu, Z.; Bian, Z.; Huang, C.; et al. Self-Repairing Tin-Based Perovskite Solar Cells with a Breakthrough Efficiency Over 11%. *Adv. Mater.* **2020**, *32* (31), 1907623.
(17) Li, W.; Li, J.; Li, J.; Fan, J.; Mai, Y.; Wang, L. Addictive-assisted construction of all-inorganic CsSnIBr2 mesoscopic perovskite solar cells with superior thermal stability up to 473 K. *J. Mater. Chem. A* **2016**, *4* (43), 17104−17110.
(18) Sanchez-Diaz, J.; Sánchez, R. S.; Masi, S.; Krecmarová, M.; Alvarez, A. O.; Barea, E. M.; Rodriguez-Romero, J.; Chirvony, V. S.; Sánchez-Royo, J. F.; Martinez-Pastor, J. P.; et al. Tin perovskite solar cells with > 1,300 h of operational stability in N2 through a synergistic chemical engineering approach. *Joule* **2022**, *6* (4), 861−883.
(19) Wang, T.; Loi, H.; Cao, J.; Qin, Z.; Guan, Z.; Xu, Y.; Cheng, H.; Li, M. G.; Lee, C.; Lu, X.; Yan, F.; et al. High Open Circuit Voltage Over 1 V Achieved in Tin-Based Perovskite Solar Cells with a 2D/3D Vertical Heterojunction. *Adv. Sci.* **2022**, *9* (18), 2200242.
(20) Wang, J.; Uddin, M. A.; Chen, B.; Ying, X.; Ni, Z.; Zhou, Y.; Li, M.; Wang, M.; Yu, Z.; Huang, J. Enhancing Photostability of Sn-Pb Perovskite Solar Cells by an Alkylammonium Pseudo-Halogen Additive. *Adv. Energy Mater.* **2023**, *13* (15), 2204115.
(21) Zhu, Z.; Jiang, X.; Yu, D.; Yu, N.; Ning, Z.; Mi, Q. Smooth and Compact FASnI3 Films for Lead-Free Perovskite Solar Cells with over 14% Efficiency. *ACS Energy Lett.* **2022**, *7* (6), 2079−2083.
(22) Yu, B.; Chen, Z.; Zhu, Y.; Wang, Y.; Han, B.; Chen, G.; Zhang, X.; Du, Z.; He, Z. Heterogeneous 2D/3D Tin-Halides Perovskite Solar Cells with Certified Conversion Efficiency Breaking 14%. *Adv. Mater.* **2021**, *33* (36), 2102055.
(23) Kayser, L. V.; Lipomi, D. J. Stretchable Conductive Polymers and Composites Based on PEDOT and PEDOT:PSS. *Adv. Mater.* **2019**, *31* (10), 1806133.
(24) Fan, X.; Nie, W.; Tsai, H.; Wang, N.; Huang, H.; Cheng, Y.; Wen, R.; Ma, L.; Yan, F.; Xia, Y. PEDOT:PSS for Flexible and Stretchable Electronics: Modifications, Strategies, and Applications. *Adv. Sci.* **2019**, *6* (19), 1900813.
(25) Liu, X.; Wang, Y.; Xie, F.; Yang, X.; Han, L. Improving the Performance of Inverted Formamidinium Tin Iodide Perovskite Solar Cells by Reducing the Energy-Level Mismatch. *ACS Energy Lett.* **2018**, *3* (5), 1116−1121.
(26) Nardes, A. M.; Kemerink, M.; de Kok, M. M.; Vinken, E.; Maturova, K.; Janssen, R. A. J. Conductivity, work function, and environmental stability of PEDOT:PSS thin films treated with sorbitol. *Org. Electron.* **2008**, *9* (5), 727−734.
(27) Onorato, A.; Invernale, M. A.; Berghorn, I. D.; Pavlik, C.; Sotzing, G. A.; Smith, M. B. Enhanced conductivity in sorbitol-treated PEDOT−PSS. Observation of an in situ cyclodehydration reaction. *Synth. Met.* **2010**, *160* (21), 2284−2289.
(28) Ramírez Quiroz, C. O.; Spyropoulos, G. D.; Salvador, M.; Roch, L. M.; Berlinghof, M.; Darío Perea, J.; Forberich, K.; Dion-Bertrand, L.; Schrenker, N. J.; Classen, A.; Gasparini, N.; Chistiakova, G.; Mews, M.; Korte, L.; Rech, B.; Li, N.; Hauke, F.; Spiecker, E.; Ameri, T.; Albrecht, S.; Abellán, G.; León, S.; Unruh, T.; Hirsch, A.; Aspuru-Guzik, A.; Brabec, C. J.; et al. Interface Molecular Engineering for Laminated Monolithic Perovskite/Silicon Tandem Solar Cells with 80.4% Fill Factor. *Adv. Funct. Mater.* **2019**, *29* (40), 1901476.
(29) Chen, J.; Zhao, X.; Cheng, Y.; Qian, J.; Wang, M.; Shen, W.; Cao, K.; Huang, Y.; Hui, W.; Gu, Y.; Chen, Y.; Gao, X.; Chen, S.; et al. Hydroxyl-Rich d-Sorbitol to Address Transport Layer/Perovskite Interfacial Issues toward Highly Efficient and Stable 2D/3D Tin-Based Perovskite Solar Cells. *Adv. Opt. Mater.* **2021**, *9* (22), 2100755.
(30) Jiang, X.; Wang, F.; Wei, Q.; Li, H.; Shang, Y.; Zhou, W.; Wang, C.; Cheng, P.; Chen, Q.; Chen, L.; Ning, Z.; et al. Ultra-high open-circuit voltage of tin perovskite solar cells via an electron transporting layer design. *Nat. Commun.* **2020**, *11* (1), 1245.
(31) Hu, F.; Chen, C.; Teng, T.; Shi, Y.; Wang, B.; Xue, D.; Xia, Y.; Chen, J.; Wang, K.; Huang, L.; Yavuz, I.; Wang, Z.; Liao, L.; et al. Constructing Charge Bridge Path for High-Performance Tin Perovskite Photovoltaics. *Adv. Energy Mater.* **2024**, *14* (10), 2302926.
(32) He, F.; Li, T.; Shen, T.; Zhao, Y.; Jin, Z.; Zhang, Z.; Pu, Y.; Deng, L.; Qin, L.; Zhan, Y.; Liu, Y.; Wang, Y.; Liang, J.; et al. Hydrophobic Electron-Transport Layer for Efficient Tin-Based Perovskite Solar Cells. *Adv. Funct. Mater.* **2024**, *34* (40), 2405611.
(33) Fang, Y.; Bi, C.; Wang, D.; Huang, J. The Functions of Fullerenes in Hybrid Perovskite Solar Cells. *ACS Energy Lett.* **2017**, *2* (4), 782−794.
(34) Zhou, J.; Hao, M.; Zhang, Y.; Ma, X.; Dong, J.; Lu, F.; Wang, J.; Wang, N.; Zhou, Y. Chemo-thermal surface dedoping for high-performance tin perovskite solar cells. *Matter* **2022**, *5* (2), 683−693.
(35) Zhou, X.; Peng, W.; Liu, Z.; Zhang, Y.; Zhang, L.; Zhang, M.; Liu, C.; Yan, L.; Wang, X.; Xu, B. Additive engineering with 2,8-dibromo-dibenzothiophene-S,S-dioxide enabled tin-based perovskite solar cells with 14.98% power conversion efficiency. *Energy Environ. Sci.* **2024**, *17* (8), 2837−2844.
(36) Said, A. A.; Aydin, E.; Ugur, E.; Xu, Z.; Deger, C.; Vishal, B.; Vlk, A.; Dally, P.; Yildirim, B. K.; Azmi, R.; Liu, J.; Jackson, E. A.; Johnson, H. M.; Gui, M.; Richter, H.; Pininti, A. R.; Bristow, H.; Babics, M.; Razzaq, A.; Mandal, S.; Allen, T. G.; Anthopoulos, T. D.;

https://doi.org/10.1021/acsami.5c05263
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

Ledinský, M.; Yavuz, I.; Rand, B. P.; De Wolf, S.; et al. Sublimed C60 for efficient and repeatable perovskite-based solar cells. *Nat. Commun.* **2024**, *15* (1), 708.
(37) Xing, Z.; Ma, S.; Chen, B.-W.; An, M.; Fan, A.; Hu, X.; Wang, Y.; Deng, L.-L.; Huang, Q.; Kanda, H.; et al. Solubilizing and stabilizing C60 with n-type polymer enables efficient inverted perovskite solar cells. *Joule* **2025**, *9*, No. 101817.
(38) Correa-Baena, J.-P.; Saliba, M.; Buonassisi, T.; Grätzel, M.; Abate, A.; Tress, W.; Hagfeldt, A. Promises and challenges of perovskite solar cells. *Science* **2017**, *358* (6364), 739−744.
(39) Bi, D.; Yi, C.; Luo, J.; Décoppet, J.-D.; Zhang, F.; Zakeeruddin, Shaik M.; Li, X.; Hagfeldt, A.; Grätzel, M. Polymer-templated nucleation and crystal growth of perovskite films for solar cells with efficiency greater than 21%. *Nat. Energy* **2016**, *1* (10), 16142.
(40) Hu, X.; Wang, H.; Wang, M.; Zang, Z. Interfacial defects passivation using fullerene-polymer mixing layer for planar-structure perovskite solar cells with negligible hysteresis. *Sol. Energy* **2020**, *206*, 816−825.
(41) Peng, J.; Khan, J. I.; Liu, W.; Ugur, E.; Duong, T.; Wu, Y.; Shen, H.; Wang, K.; Dang, H.; Aydin, E.; Yang, X.; Wan, Y.; Weber, K. J.; Catchpole, K. R.; Laquai, F.; De Wolf, S.; White, T. P.; et al. A Universal Double-Side Passivation for High Open-Circuit Voltage in Perovskite Solar Cells: Role of Carbonyl Groups in Poly(methyl methacrylate). *Adv. Energy Mater.* **2018**, *8* (30), 1801208.
(42) Zhu, Z.; Mi, Q. Substituted thiourea as versatile ligands for crystallization control and surface passivation of tin-based perovskite. *Cell Rep. Phys. Sci.* **2022**, *3* (1), No. 100690.
(43) He, Y.; Chen, H.-Y.; Hou, J.; Li, Y. Indene−C60 Bisadduct: A New Acceptor for High-Performance Polymer Solar Cells. *J. Am. Chem. Soc.* **2010**, *132* (4), 1377−1382.
(44) Kang, H.; Cho, C.-H.; Cho, H.-H.; Kang, T. E.; Kim, H. J.; Kim, K.-H.; Yoon, S. C.; Kim, B. J. Controlling Number of Indene Solubilizing Groups in Multiadduct Fullerenes for Tuning Optoelectronic Properties and Open-Circuit Voltage in Organic Solar Cells. *ACS Appl. Mater. Int.* **2012**, *4* (1), 110−116.

https://doi.org/10.1021/acsami.5c05263
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX